DATA ANALYTICS PROJECT PLAN // POLYGON ON-CHAIN INTELLIGENCE

Polymarket Whale Consensus & Smart Money Analytics

Identifying highly profitable wallets on Polygon to determine if specialized traders dictate prediction market outcomes.

01 // DOMAIN & APPLICATION

Project Objective & Core Hypothesis

PRIMARY OBJECTIVE

Do Whales Dictate Market Consensus?

Identify specialized high-profit wallets on Polymarket and quantify how strongly their early positions drive final market resolution.

Prediction Markets: Financial contracts settling on binary event outcomes.

Information Asymmetry: Informed traders deploy capital before public consensus forms.

On-Chain Transparency: Every order book fill settles publicly on Polygon.

Signal Extraction: Isolate high-conviction alpha from retail trading noise.

02 // EMPIRICAL CONTEXT

Capital Concentration on Polymarket

50%

Total volume driven by professional whales

0.4%

Volume share from bottom 70% retail users

03 // ARCHITECTURAL BLUEPRINT

End-to-End Data Science Pipeline

01

Data Source

Polygon RPC nodes and Dune SQL tables.

02

Ingestion

Decode CTF Exchange order fill logs.

03

Engineering

Compute wallet P&L and entry timing.

04

Computation

Iterative K-Means and DBSCAN clustering.

05

Deliverables

Smart money classifier and network graph.

04 // WORK ESTIMATION & STRUCTURE

Analytics Project Planning Roadmap

01

Domain Focus

Scope & Application

Target Market

Prediction market microstructure on Polygon network.

02

Data Sources

On-Chain Availability

Infrastructure

Public smart contracts and indexed Dune Spellbooks.

03

Work Allocation

Estimation & Breakdown

60% Engineering
40% ML Modeling

- Data Ingestion
- Data Engineering
- Analytics Computation
- Data Visualization

04

Target Output

Core Deliverables

System Outcome

Automated whale flagging and consensus tracker.

05 // DATA AVAILABILITY

On-Chain Infrastructure & Data Sources



Polygon RPC Nodes

Raw block receipts via Web3 API.



Polymarket Contracts

CTF Exchange order settlement logs.



Dune Analytics

Indexed trade history SQL tables.

06 // ESSENTIAL WORK: INGESTION & ENGINEERING

From Raw Polygon Logs to Wallet Features

01

Event Extraction

Query OrderFilled smart contract events.

02

Deduplication

Filter taker-side volume against double-counting.

03

Proxy Mapping

Link CLOB proxy wallets to traders.

04

Feature Vectors

Calculate win rate and entry lead-time.

07 // ESSENTIAL WORK: ANALYTICS COMPUTATION

Unsupervised Wallet Clustering Algorithms

PARTITION CLUSTERING

K-Means Segmentation

Centroid Grouping: Segments wallets into k=5 behavioral tiers.

Core Features: Log USDC volume and historical win rate.

Strength: Clean boundary between retail and mid-tier traders.

DENSITY-BASED CLUSTERING

DBSCAN Anomaly Detection

Outlier Isolation: Flags extreme high-profit whale anomalies.

Noise Filtering: Separates high-frequency wash bots from humans.

Strength: Discovers non-linear syndicates without preset k.

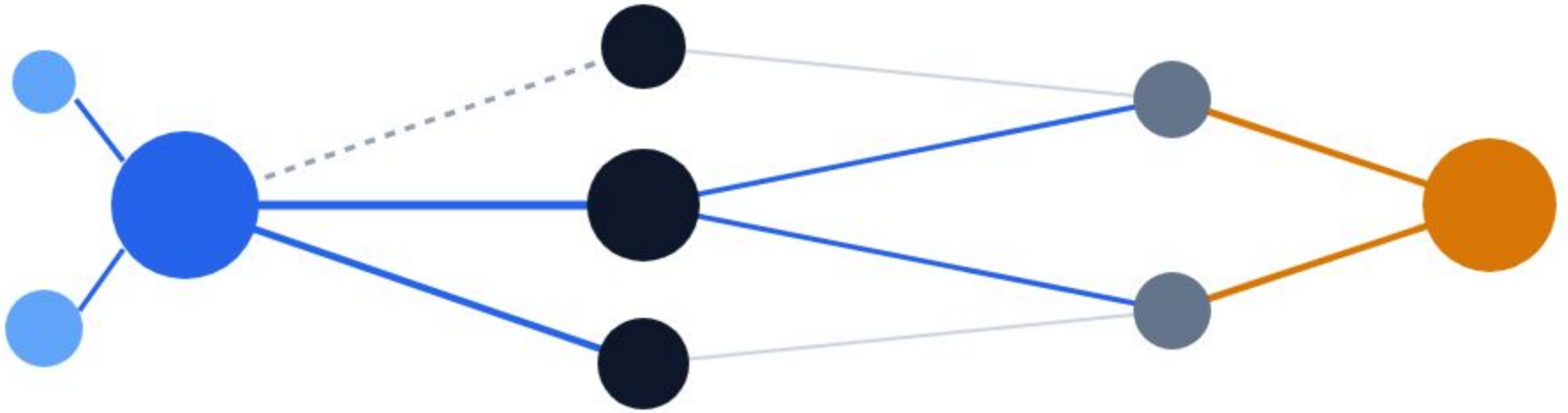
08 // CLASSIFICATION TAXONOMY

Identified Wallet Clusters & Behavioral Profiles

Cluster Archetype	Trade Volume	Win Rate	Entry Timing	Consensus Role
Smart Money Whales	> $100,000	68% – 78%	First 15% of lifecycle	**Primary Price Setter**
Event Snipers	$50,000+ burst	74% – 85%	News catalyst spikes	Rapid Odds Shifter
HFT Arbitrage Bots	> $250,000	51% – 54%	Continuous (>50 tx/day)	Liquidity Spread Neutral
Mid-Tier Dolphins	$5,000 – $50,000	52% – 58%	Mid-market trend	Momentum Follower
Retail Participants	< $1,000	38% – 46%	Late consensus phase	Exit Liquidity

09 // DELIVERABLE I: DATA VISUALIZATION

Smart Money Co-Trading Network Graph



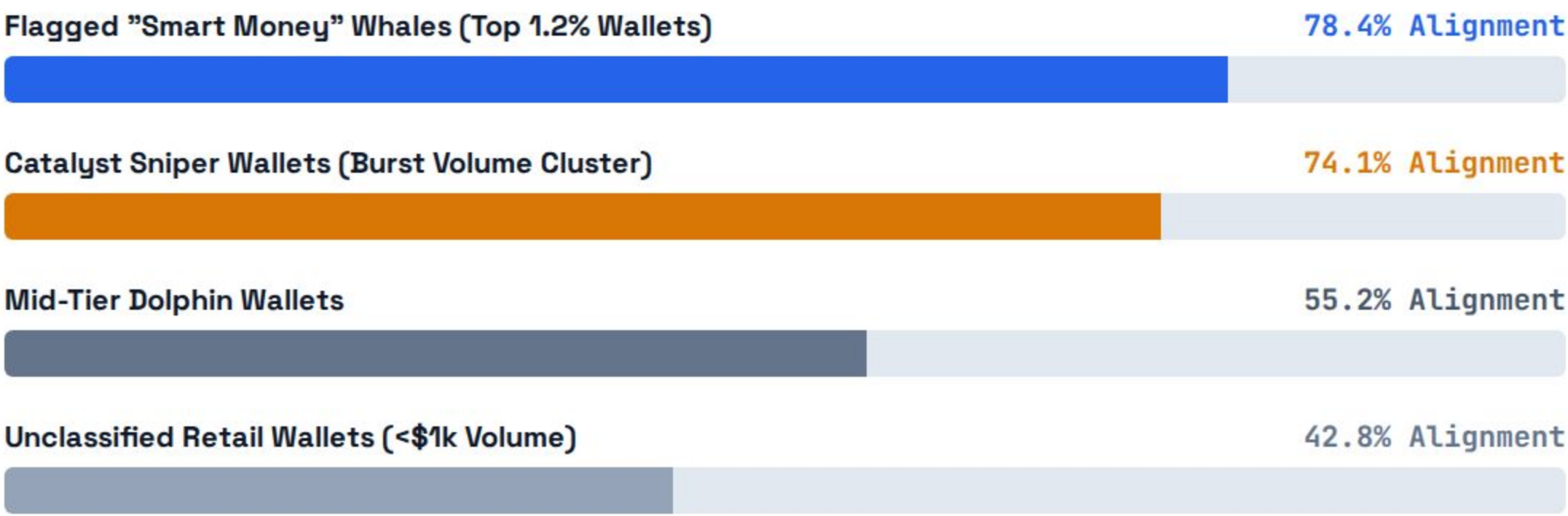
STAGE 1: ALPHA ENTRY
Lead Whale Wallets (T+0h)

STAGE 2: SYNDICATE ECHO
Copy-Trading Clusters (T+2h)

STAGE 3: ORDER BOOK SHIFT
Retail Momentum (T+12h)

STAGE 4: RESOLUTION
Final Market Consensus

Initial Position Alignment with Final Resolution



PROJECT DELIVERABLES SUMMARY

Small Whale Subsets Anchor Market Consensus

Combining Polygon RPC ingestion, DBSCAN clustering, and network graph classification flags high-conviction smart money in real time.